Nationwide France Apartment Valuation

Phase 4 Data Engineering, Leakage Controls and Chronological Model Benchmark

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Protocol and pre-benchmark edition

Abstract

This protocol defines a nationwide France apartment automated valuation benchmark using annual geolocated DVF transactions from 2021–2025. It specifies the cohort, global spatial representation, strictly lagged comparable-sale features, hierarchical baseline, CatBoost candidates, chronological split, paired uncertainty analysis and territorial diagnostics before national model results are available. This edition intentionally reports no accuracy number; rerunning the generator after the benchmark replaces pending sections with audited empirical tables.

Keywords: automated valuation model; Geo-DVF; France; CatBoost; historical comparable sales; chronological validation; regional robustness.

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1 Executive assessment

Phase 4 extends the earlier Paris research into one national apartment model. The central design choice is comparability rather than superficial feature parity: Paris-only transport, school, park, noise and building snapshots are not copied into other regions without a nationally uniform source contract. The national model instead uses DVF property attributes, transaction time, hierarchical territorial identity, global geometry and point-in-time comparable-sale evidence.

Protocol edition. The national Gold table and fitted model are not present in this workspace. No accuracy number is asserted. The manuscript will be regenerated from the frozen benchmark artifacts without manual score entry.

Decision status

- The data, benchmark and reporting protocols are fully specified and tested.
- National accuracy may be claimed only after the 2025 frozen-test artifacts exist.
- Regional MAE must be interpreted beside MdAPE, hit rate and test-row count.
- The output is an automated indication, not a certified appraisal.

2 Research objective and contributions

The target is total transaction value in euros for an ordinary apartment sale. The study asks whether one model can learn heterogeneous French markets while preserving local evidence and honest temporal evaluation. Its contributions are:

1. chunk-safe processing of the national annual Geo-DVF archives;
2. explicit support for Corsican and overseas territorial identifiers;
3. spherical H3 and Cartesian location features valid outside Lambert-93;
4. local comparables subject to a 90-day information-availability lag;
5. a training-only hierarchical postcode-to-national median baseline;
6. a 2021–2023/2024/2025 chronological benchmark with paired bootstrap uncertainty;
7. region and sufficiently sized department diagnostics;
8. atomic artifacts, checksums, a reusable production bundle and inference CLI.

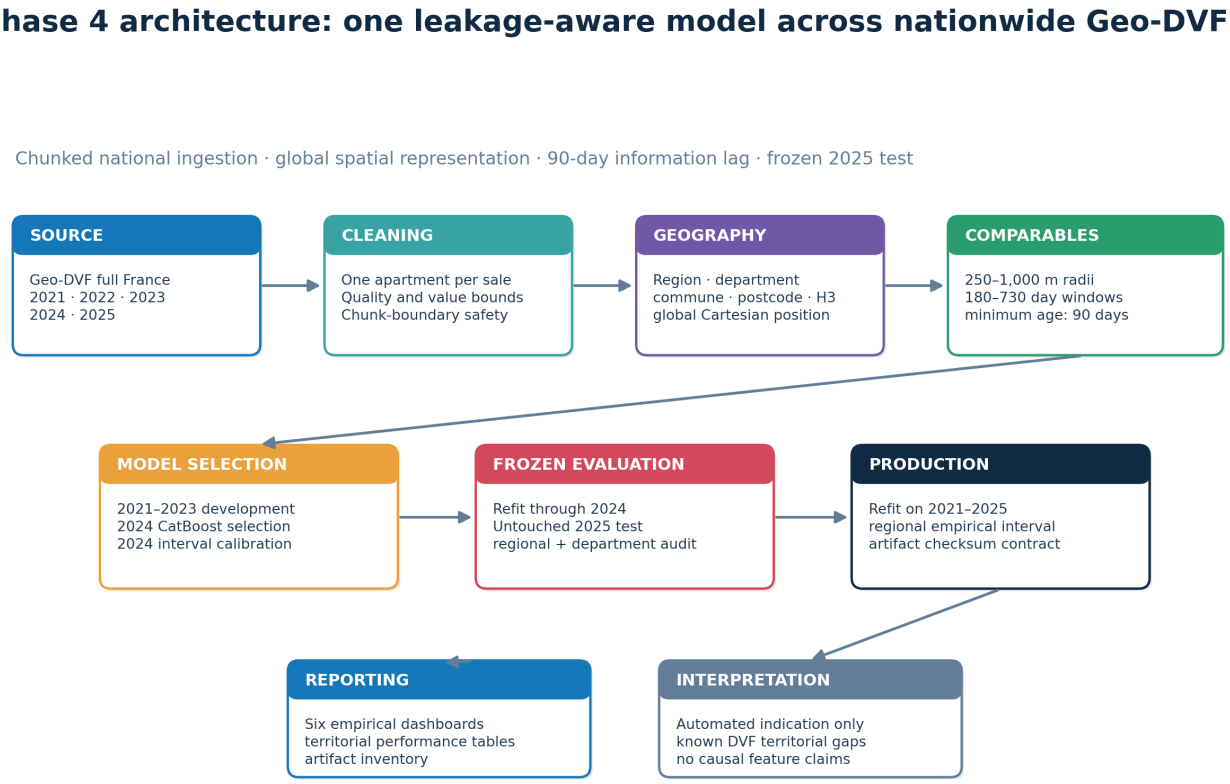


Figure 1: Phase 4 source-to-report architecture and chronological evaluation design.

3 Data source and analytical cohort

The source is the normalized and geolocated Geo-DVF annual full archive for 2021–2025. CSV files are streamed in bounded chunks. The last mutation in each chunk is carried into the next chunk so a multi-line transaction is never classified before all its property rows are available.

The cohort retains *Vente* mutations containing exactly one apartment and no house or commercial premises. Dependencies such as a cellar or parking place may remain bundled in the mutation consideration. Required fields are date, transaction value, apartment surface, commune and valid WGS84 coordinates. Broad national quality bounds are 9–300 m^2, EUR 10,000–10,000,000 and EUR 100–30,000 per m^2.

Territorial row counts are pending the national Gold feature build.

Geo-DVF contains no open observations for departments 57, 67, 68, 976. “Nationwide” therefore means every territory represented upstream, not every French department without exception.

4 Feature engineering contract

The fitted contract contains 46 inputs: 39 numeric and 7 categorical variables. The current transaction’s target and price per square metre are excluded.

Table 1: Phase 4 feature families.

Family	Inputs	Role
Property	7	Surface, rooms, lots, terrain and scale/density transforms
Calendar	4	Seasonality and elapsed market time
Global location	5	WGS84 and unit-sphere Cartesian coordinates
Territorial identity	7	Region, department, commune, postcode and H3 r7/r8/r9
Historical comparables	23	Lagged counts, medians, dispersion, similarity, age, distance and momentum

4.1 Global spatial representation

Lambert-93 is unsuitable for overseas departments. Longitude λ and latitude ϕ are therefore supplemented by unit-sphere coordinates

$$(x, y, z) = (\cos\phi\cos\lambda, \cos\phi\sin\lambda, \sin\phi), \tag{1}$$

and H3 cells. Nearby transactions are queried using haversine distance within each department.

4.2 Leakage-safe comparable sales

For target sale i at date t_i, candidate j is eligible only if

$$90 \le t_i - t_j \le 730 \quad \text{days}, \qquad d(i,j) \le 1{,}000 \text{ m}. \tag{2}$$

Windows at 180, 365 and 730 days and radii at 250, 500 and 1,000 metres produce counts and median prices per square metre. Exponential recency/distance weights, surface and room similarity, effective sample size, dispersion and momentum summarize local evidence. The current transaction cannot become its own input.

Table 2: Primary leakage and integrity controls.

Risk	Enforced control
Mutation target duplication	One row only after transaction-level local-type audit
Future comparable sales	Minimum 90-day age and positive chronological difference
Target-derived leakage	Current value and price/m^2 excluded from fitted inputs
Geographic projection failure	Haversine distances and global unit-sphere coordinates
Model-selection leakage	2024 selection separated from untouched 2025 evaluation
Partial artifact corruption	Sibling `.part` writes followed by atomic promotion
Silent territorial overclaim	Upstream missing departments recorded in every report

5 Benchmark design

Development uses 2021–2023. Two predeclared CatBoost configurations are compared on 2024; no 2025 outcome participates in selection. The winner is refitted through 2024 and evaluated once on 2025. Only after evaluation is the production model refitted on all 2021–2025 observations.

Table 3: Predeclared CatBoost candidates for 2024 model selection.

Candidate	Depth	Loss	Regularization
1	8	MAE	$\ell_2 = 8$
2	9	Huber ($\delta = 1$)	$\ell_2 = 10$

The objective is fitted to $\log(1+y)$ and transformed back to non-negative euros. The benchmark baseline calculates median historical price per square metre by postcode and falls back successively to commune, department, region and national median. Every lookup is learned exclusively from the corresponding training period.

5.1 Metrics and paired uncertainty

The report publishes MAE, RMSE, R^2, MAPE, median absolute percentage error and hit rates within $\pm 10\%$ and $\pm 20\%$. Because model and baseline predict the same transactions, uncertainty is estimated from paired absolute-error differences. A deterministic bootstrap uses 300 repetitions and at most 250,000 test rows to bound memory. The exact sample size is stored in the benchmark JSON.

6 Frozen national benchmark

Benchmark pending. The national Gold table and fitted model are not present in this workspace. No accuracy number is asserted in this edition. The registered comparison is Phase 4 CatBoost versus the hierarchical training-only geographic median on the identical untouched 2025 transactions.

7 Territorial robustness

Raw euro MAE is expected to be higher in expensive markets. Regional assessment therefore requires MdAPE, hit rates, R^2 and sample size beside MAE. Department rows require at least 100 test transactions; small segments are suppressed rather than presented as stable estimates.

Regional and department metrics are pending the frozen 2025 evaluation. The reporting contract requires at least 100 test sales before a territorial row is published and at least 500 calibration sales before a region receives its own empirical interval ratios.

8 Calibration and uncertainty

The empirical 90% interval uses 2024 target-to-prediction ratios. A region gets its own calibration ratios only with at least 500 observations; otherwise it inherits the national interval. Observed 2025 coverage evaluates temporal transfer but is not a formal conditional guarantee for a specific apartment.

9 Visual diagnostic suite

Figures are generated only from existing artifacts. Missing benchmark outputs produce no placeholder chart and no synthetic metric. Feature importance records predictive usage rather than a causal housing premium.

10 Reproducibility and artifact inventory

Artifact	Status	Bytes
Source manifest	**pending**	—
Gold feature table	**pending**	—
Feature quality report	**pending**	—
Benchmark JSON	**pending**	—
Validation trials	**pending**	—
Benchmark comparison	**pending**	—
Frozen predictions	**pending**	—
Regional metrics	**pending**	—
Department metrics	**pending**	—
Feature importance	**pending**	—
Production model	**pending**	—

```
python -m paris_avm.data.acquire_phase4
python -m paris_avm.features.phase4
python -m paris_avm.modeling.benchmark_phase4
python -m paris_avm.visualization.phase4
python -m paris_avm.reporting.phase4
powershell -ExecutionPolicy Bypass -File .\docs\paper\compile_phase4.ps1
```

11 Limitations and responsible use

- DVF omits condition, renovation, floor, elevator, view and occupancy.
- Mutation value can bundle dependencies or transaction-specific rights.
- Departments 57, 67, 68 and 976 are absent from the source.
- Sparse submarkets may have weak comparable coverage and unstable errors.
- Administrative/H3 identity is predictive context, not a causal estimate.
- A 2025 holdout does not establish performance in 2026 or later markets.
- The interval is empirical and marginal, not a certified property guarantee.
- Predictions are not notarial valuations, lending decisions or investment advice.

12 Conclusion

Phase 4 defines a scalable and auditable path from a Paris research model to a national France AVM. Its strongest contribution is the combination of global geography, local point-in-time market evidence, chronological evaluation and territorial reporting. In this protocol and pre-benchmark edition, claims remain limited to the artifacts actually present. The empirical edition is generated by the identical source after the full national benchmark completes.

A Machine-readable deliverables

The benchmark publishes JSON metrics, validation trials, a two-method comparison, regional and department CSV tables, Parquet predictions with paired errors, feature importance and a checksummed Joblib production bundle. The Markdown and LaTeX reports consume these outputs without refitting or selecting a model.

B Source references

- Geo-DVF dataset catalogue on data.gouv.fr
- Geo-DVF annual file index
- Open Licence 2.0